Check for updates

Health Science Reports

WILEY

Health Science Reports
Open Access

ORIGINAL RESEARCH OPEN ACCESS

Prevalence and Temporal Trends of Transfusion-Transmissible Infections Among Blood Donors in a Municipal Hospital in Navrongo, Ghana: A Retrospective Study (2018–2024)

Francis Yennube Duut[1] | Samuel Gana Kolog[1] | Francis Kwame Ayimavor[1] | Evans Anuah[1] | Courage Honorfe[2] | Ramzy Bugri Anafo[2] | Godfred Agongo[1] | James Abugri[1] | Auphedeous Yinme Dang-I[3] | Sylvester Donne Dassah[1]

[1]Department of Biochemistry and Forensic Sciences, School of Chemical and Biochemical Sciences, University for Technology and Applied Sciences, Navrongo, Upper East Region, Ghana | [2]Laboratory Unit, War Memorial Hospital, Ghana Health Service, Navrongo, Upper East Region, Ghana | [3]Department of Applied Chemistry, School of Chemical and Biochemical Sciences, University for Technology and Applied Sciences, Navrongo, Upper East Region, Ghana

Correspondence: Sylvester Donne Dassah (sddassah@cktutas.edu.gh)

Received: 22 September 2025 | **Revised:** 1 February 2026 | **Accepted:** 2 March 2026

Keywords: blood donors | HBV | HCV | HIV | syphilis | transfusion-transmissible infections (TTIs)

ABSTRACT

Background and Aim: Blood transfusion is a vital and life-saving procedure, saving millions of lives across the world every year. However, it also presents a significant risk of spreading transfusion-transmissible infections (TTIs) such as hepatitis B virus (HBV), hepatitis C virus (HCV), human immunodeficiency virus (HIV), and syphilis (*Treponema pallidum*). This study aimed to determine the seroprevalence and temporal trends of TTIs among blood donors at a primary healthcare hospital in the Upper East Region of Ghana.

Methods: A retrospective analysis of blood donation screening records from War Memorial Hospital, Navrongo, was conducted for the period January 2018 to December 2024. Data on HBV, HCV, HIV-1/2, and syphilis were extracted and analysed using descriptive statistics to determine seroprevalence and temporal trends across demographic groups.

Results: A total of 5741 blood donation records were analysed, with males 98.1% of donors. The overall seroprevalence of TTIs was 18.2%. Individual seroprevalence rates for HBV, HCV, *syphilis*, and HIV-1/2 infections were 10.1%, 6.5%, 2.1%, and 0.5%, respectively. A decline in both donor numbers and TTIs seroprevalence was observed during the peak COVID-19 period (2020–2021), followed by a gradual recovery in donor numbers in 2022–2023. HBV seroprevalence decreased from 15.8% in 2018 to 7.1% in 2021. HCV seroprevalence was highest in 2023 (8.1%) and lowest in 2020 (5.1%). Syphilis seroprevalence showed a general downward trend with a temporal increase in 2022.

Conclusion: TTIs remain a public health concern for blood donations in the area, particularly HBV and HCV. Although declining trends were observed, the fluctuations highlight the need to strengthen screening protocols and improved criteria for donor selection. The low seroprevalence of HIV and syphilis is comforting but requires continued preventive efforts to maintain the progress.

Abbreviations: CHPS, Community-based Health Planning and Services; CI, Confidence interval; COVID-19, Coronavirus disease 2019; ELISA, Enzyme-linked immunosorbent assay; HBV, hepatitis B virus; HCV, hepatitis C virus; HDSS, Health and Demographic Surveillance System; HIV, human immunodeficiency virus; KNM, Kassena Nankana Municipal; OR, Odds ratio; TTI, Transfusion-transmissible infection; WHO, World Health Organization.

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *Health Science Reports* published by Wiley Periodicals LLC.

Summary

- Blood transfusion presents the risk of transfusion-transmissible infections (TTIs). Seroprevalence of TTI of 21.0% has been reported across a study in four hospitals in Ghana, with HBV (6.6%), HCV (4.9%), HIV (2.9%), and *T. pallidum* (6.8%).
- This study observed seroprevalence of 18.2% for TTIs, with HBV (10.1%), HCV (6.5%), *T. pallidum* (2.1%), and HIV (0.5%).
- TTIs remain a major risk among blood donors in the Upper East Region of Ghana, with a notably high seroprevalence of HBV and HCV. The persistent high burden of these TTIs calls for improvement and strict adherence to blood screening protocols.

1 | Introduction

Blood transfusion is a vital and life-saving procedure, saving millions of lives across the world every year [1]. However, it also presents the risk of transfusion-transmissible infections (TTIs) such as hepatitis B virus (HBV), hepatitis C virus (HCV), human immunodeficiency virus (HIV), and syphilis (*Treponema pallidum* infection), particularly in resource-limited settings where comprehensive donor screening is a challenge [2–7]. Globally, annual blood donations are estimated to be about 118.5 million [1]. Therefore, ensuring a safe blood supply through comprehensive donor blood screening is essential for patient care and for the broader control of viral hepatitis. Screening of donated blood for TTIs such as HBV and HCV has become standard practice in Ghana. However, residual risks remain due to factors such as window periods of incubation, prolonged phases of viremia, high infection rate, occult infection, and variations in test sensitivity [8–12].

Ghana has been identified as having intermediate to high endemicity of viral hepatitis due to HBV and HCV, with transmission occurring through various routes, including mother-to-child transmission, unsafe medical practices, and blood transfusion [5, 10, 13–15]. The World Health Organization's Global Health Sector Strategy on Viral Hepatitis aims to eliminate viral hepatitis as a public health threat by 2030, targeting a 90% reduction in new infections and a 65% reduction in mortality [16]. Achieving these goals requires comprehensive surveillance systems, effective prevention strategies, and improved access to testing and treatment, particularly in high-burden regions like sub-Saharan Africa. Previous studies in Ghana have reported HBV surface antigen (HBsAg) seroprevalence ranging from 6% to 15% among various populations, with significant regional variations [17, 18]. HCV seroprevalence, on the other hand, has been reported to range from 1% to 8%, depending on the population studied and geographical location [19, 20]. The northern regions of Ghana, including the Upper East Region, have historically experienced higher seroprevalence of viral hepatitis compared to southern regions [21]. The disparity has been attributed to factors including low socioeconomic status, limited access to healthcare services, and cultural practices [5, 22]. In addition to HBV and HCV, HIV and syphilis remain major TTIs of major concern in Ghana. HIV seroprevalence among blood donors has declined over the years but continues to pose a residual risk due to early infection during the serological window period [23–25]. Syphilis, caused by *Treponema pallidum*, also contributes significantly to donor deferral rate and transfusion safety concerns, with seroprevalence varying across regions [26].

Ghana's blood transfusion service operates through a network of hospital-based blood banks and a few regional blood centers, coordinated by the National Blood Service [27]. The country has made significant progress in improving blood safety through the implementation of mandatory screening for major TTIs, including HBV, HCV, HIV, and syphilis [28]. However, challenges remain in ensuring universal coverage and maintaining consistent quality standards across all facilities.

Despite the recognized burden of TTIs in Ghana, particularly in rural settings such as the Upper East Region, there is the absence of studies that have systematically examined the seroprevalence and temporal trends of TTIs among blood donors in this setting. Blood donors in this region might represent a relatively healthy subset of the population, yet their infection rates can provide valuable insights into community-level disease burden and transmission dynamics [4, 5]. Understanding the temporal patterns of TTIs among blood donors can contribute to broader surveillance efforts and support the development of targeted interventions. This retrospective study sought to determine the seroprevalence and trends of TTIs among blood donors at a primary healthcare hospital in the Upper East Region of Ghana. To the best of our knowledge, this study is the first long-term (5-year) trend analysis of the seroprevalence of HBV and HCV and other TTIs in the Upper East Region of Ghana. The findings from this study contribute to the epidemiology of TTIs in rural Ghana and provide valuable data needed to strengthen blood safety protocols and donation management strategies, guide public health interventions, and inform ongoing efforts to achieve the World Health Organization (WHO) target of eliminating viral hepatitis as a public health threat by 2030.

2 | Materials and Methods

2.1 | Study Setting

This study was conducted at the War Memorial Hospital in Navrongo in the Kassena Nankana Municipal (KNM) in the Upper East Region of Ghana. The KNM lies within the Guinea savannah woodland ecological zone (approximately between latitude 11°10′ and 10°3′ North and longitude 10°1′ West), shares a boundary with Burkina Faso to the north, and is characterized by a unimodal rainfall pattern from May to September and a prolonged dry season dominated by harmattan winds [29]. The municipality is predominantly rural and has an estimated population of 99,895 inhabitants [30]. Livelihoods are primarily agrarian, with subsistence farming, along with livestock rearing, and small-scale fishing.

Healthcare delivery in the municipality is provided through a network of facilities, including the municipal hospital (i.e., War Memorial Hospital), health centers, and Community-based Health Planning and Services (CHPS) compounds, in addition to the University of Technology and Applied Sciences hospital, private clinics, and informal providers such as traditional

23988835, 2026, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/hsr2.72122 by EBMG ACCESS - GHANA, Wiley Online Library on [23/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

healers and chemical sellers. The municipality also has the Navrongo Health Research Center, which operates the Navrongo Health and Demographic Surveillance System (HDSS), an important platform for monitoring health and demographic trends in the area [31]. This health system context is important for understanding blood donor patterns and the burden of TTIs in the municipality.

The War Memorial Hospital is a first-level referral health facility in the KNM and has outpatient and inpatient care. These include surgical and obstetric care, public health services, as well as laboratory services. Blood transfusion is housed under laboratory services [32]. It is a 123-bed facility that receives referrals from the primary level, including private clinics, as well as from neighboring districts and towns in Burkina Faso. According to hospital records, approximately 55,000 people visit the outpatient department on average each year, and 10,000 patients are admitted for inpatient care. Blood for transfusion services in the hospital is screened for transfusion-transmissible infections using serological rapid diagnostic tests, which ensures efficiency and prompt service delivery.

2.2 | Ethical Consideration

Ethical approval for the conduct of this study was obtained from the Navrongo Health Research Center Institutional Review Board (Approval ID: NHRCIRB535). Written permission was obtained from the hospital authority prior to the start of study activities. Informed consent from participants was waived by the ethics committee since the study involved a review of previously collected data. The data retrieved from the records were assigned unique study codes without donor identifiers to ensure confidentiality. All the study procedures were implemented in accordance with the Declaration of Helsinki.

2.3 | Study Design and Records Selection

This was a retrospective cross-sectional study that reviewed and analysed archived data from the blood donation registry at the War Memorial Hospital blood bank in Navrongo between the period of January 1, 2018, and December 31, 2024 excluding 2019 due to incomplete data. A 5-year period was chosen to ensure sufficient sample size, capture temporal trends including the COVID-19 period, and provide recent, complete, and policy-relevant data. War Memorial Hospital was purposively selected because it is the only facility within the KNM that provides blood transfusion services and had the required data for the implementation of this study within the municipality. Both voluntary and replacement donors who visited the hospital's blood bank within the study period constituted the study population. Records of blood donations were retrieved from paper files and checked for completeness in terms of basic socio-demographic information (i.e., age, sex, and year) and screening information (i.e., blood group type and availability of laboratory test results on HBV, HCV, HIV-1/2, and syphilis (*T. pallidum* infection)). Blood donation procedures included pre-donation education, administration of a donation history questionnaire, and age and hemoglobin requirements, followed by testing for TTIs in the facility, donor blood screening for TTIs was based on serology using rapid diagnostic immunochromatographic tests. Information on the specific RDT brands used (2018–2024) was not available in the source records, which limited our ability to report kit-specific sensitivity and specificity. However, RDTs for TTIs such as HIV, HBV, HCV, and syphilis used in similar settings typically report sensitivity ranging from approximately 90% to 99% and specificity from 95% to 99% [33, 34].

The sample size for the study was based on available blood donation records screened during the 5-year review period.

2.4 | Data Analysis and Presentation

The retrieved data was checked for inconsistencies and completeness and entered into a predesigned Microsoft™ Excel spreadsheet (MS Office Excel® 2021). The data was exported into GraphPad Prism version 9.0 (San Diego, CA, USA) and STATA version 17.0 software (STATA Corporation, Texas, USA) for analysis. The dependent variables were HBV, HCV, HIV-1/2, and syphilis (*T. pallidum* infection) test results (coded as negative or positive). Each record represents a single donation rather than a unique donor. Differences in the total sample size and the case counts for HBV, HCV, HIV-1/2, and syphilis were due to incomplete or missing laboratory results for some records. Data from 2019 was not included for the analysis because the records were incomplete. All incomplete or missing data were excluded listwise. Descriptive statistics were used to describe HBV, HCV, HIV-1/2, and syphilis seroprevalence among various demographic groups. ABO-Rh-specific seroprevalence of HBV and HCV was conducted to explore potential correlates with ABO blood groups. Pearson's chi-square test was used to compare differences between categorical variables. A p-value less than 0.05 was considered to be statistically significant.

3 | Results

3.1 | General Characteristics of the Blood Donor Population

A total of 5741 records were retrieved from the blood bank registry for analysis in this study, which constituted the sample size. The average age of the donors was 30.54 ± 7.40 standard deviations (range: 16–68 years). The blood donation population was predominantly males (98.1%), with only 1.9% being female donors (Table 1). The majority of the blood donors were in the age groups of 20–30 (51.7%) and 31–40 (35.2%). Fewer donations came from donors who were below 20 years (3.0%) or above 50 years (1.2%). The highest number of blood donations was recorded in 2023 (24.1%), post the COVID-19 pandemic era, while the least number of blood donations (8.8%) was recorded in 2020, a COVID-19 pandemic year. A noticeable decline in the donation numbers was observed during the peak period of the COVID-19 pandemic (2020–2021), followed by a gradual recovery in 2022–2023.

3.2 | ABO Rhesus Blood Group Distribution of the Blood Donor Population

The most common blood group was O-rhesus positive, 54.4%, and the least was the AB-negative blood group, 0.3% (Figure 1).

23988835, 2026, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/hsr2.72122 by EBMG ACCESS - GHANA, Wiley Online Library on [23/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Demographic characteristics of the blood donors.

Variables	Frequency	Percent
Mean age ± SD (years)	30.54 ± 7.40	
Sex		
Male	5623	98.1
Female	112	1.9
Age group (years)		
< 20	159	3.0
20–30	2751	51.7
31–40	1874	35.2
41–50	470	8.8
> 50	65	1.2
Year		
2018	990	17.2
2020	506	8.8
2021	949	16.5
2022	808	14.1
2023	1386	24.1
2024	1102	19.2
Total	5741	100.0

Abbreviation: SD, standard deviation.

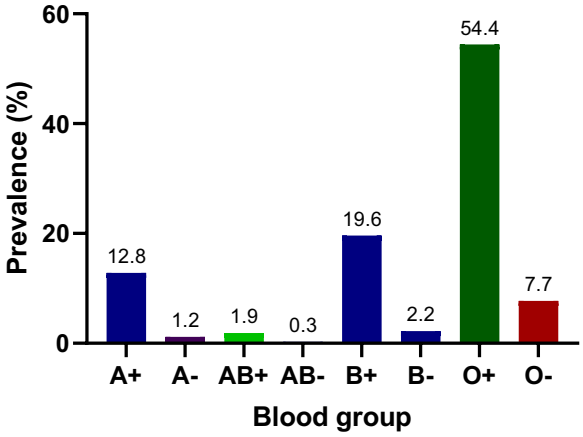


FIGURE 1 | Distribution of ABO Rhesus factor blood groups among donors.

Seroprevalence of hepatitis B and C virus among blood donors

The seroprevalence of HBV and HCV among the blood donors was 10.1% and 6.5%, respectively (Table 2). The seroprevalence of hepatitis B and C infection among blood donors stratified by gender indicated that male donors had the highest seroprevalence (10.2%) compared to females (3.3%) for HBV and 6.6% of males compared to 1.7% of females for HCV, respectively.

The seroprevalence of HBV varied significantly across age groups ($p = 0.002$), ranging from 11.7% in the age group 20–30 years to 5.7% in the age group 41–50 years (Table 2). HCV seroprevalence ranged from 10.5% in the above 50-year-old age group to 5.2% in the 31–40-year age group.

No significant difference or particular trend was observed with the distribution of the seroprevalence of hepatitis B and C infection among the various ABO-rhesus blood groups (Supplementary Figure 1). These findings are presented for descriptive purposes only. The highest hepatitis B virus seroprevalence was observed in blood groups B+ (11.1%) and B− (11.1%), and the least was recorded among blood group O− (6.1%). With the hepatitis C virus, the highest seroprevalence was observed in blood group AB+ (13.0%), and the least was 0.0%, observed among blood group AB−.

3.3 | Seroprevalence of Other Transfusion-Transmissible Infections Among Blood Donors

The analysis of the other transfusion-transmissible infections, such as syphilis and HIV-1/2, which are often screened for during blood donation services, showed that the seroprevalence of syphilis was 2.1% and the seroprevalence of HIV-1/2 was 0.5% among the donors (Table 3). No positive donors were reported for both HIV and syphilis among female donors. Seroprevalence for both HIV and syphilis was low but fluctuated among the various age groups, with the highest proportion observed in the age group > 50 years.

3.4 | Coinfection Patterns of Transfusion-Transmissible Infection Across Blood Donations

The distribution of transfusion-transmissible infection coinfections among screened blood donations is summarized in Figure 2. The most frequently observed coinfection was HBV and HCV, with a seroprevalence of 4.6%. This was followed by HBV and syphilis coinfections (4.1%). Coinfections

23988835, 2026, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/hsr2.72122 by EBMG ACCESS - GHANA, Wiley Online Library on [23/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Seroprevalence of hepatitis B and C infection among blood donors stratified by sex and age group.

Variables	HBV			HCV		
	Total *N*	Positive *n* (%)	*p* value	Total *N*	Positive *n* (%)	*p* value
Sex						
Male	3885	397 (10.2)	0.074	3732	245 (6.6)	0.13
Female	61	2 (3.3)		59	1 (1.7)	
Age group (years)						
< 20	94	8 (8.5)	0.002	91	6 (6.6)	0.19
20–30	1923	225 (11.7)		1843	134 (7.3)	
31–40	1305	110 (8.4)		1260	66 (5.2)	
41–50	316	18 (5.7)		308	19 (6.2)	
> 50	37	3 (8.1)		38	4 (10.5)	
Total	3946	399 (10.1)		3791	246 (6.5)	

Note: The table shows the total number of records (N), the number that were positive, (n) and the percentage (%) for blood donors screened for either HBV or HCV. Percentages are calculated based on the total number of blood donor records within each group. *P*-values were obtained from Pearson's chi-square test of differences between categorical variables within each group. Note: The totals for HBV and HCV do not equal the overall records reviewed (5741) because some records had missing laboratory results.

TABLE 3 | Seroprevalence of syphilis and HIV among blood donors.

Variables	HIV			Syphilis		
	Total *N*	Positive *n* (%)	*p* value	Total *N*	Positive *n* (%)	*p* value
Sex						
Male	3628	20 (0.6)	0.564	3624	78 (2.2)	0.25
Female	60	0 (0.00)		60	0 (0.0)	
Total	3688	20 (0.5)		3684	78 (2.1)	
Age group (years)						
< 20	88	1 (1.1)	0.133	88	1 (1.1)	0.63
20–30	1799	8 (0.4)		1794	43 (2.4)	
31–40	1227	3 (0.2)		1230	21 (1.7)	
41–50	298	1 (0.3)		297	8 (2.7)	
> 50	36	1 (2.8)		37	1 (2.7)	

Note: The table shows the total number of records (N), the number that were positive, (*n*) and the percentage (%) for blood donors screened for either HIV-1/2 or syphilis. Percentages are calculated based on the total number of blood donor records within each group. *P*-values were obtained from Pearson's chi-square test of differences between categorical variables within each group. Note: The totals for HIV-1/2 and syphilis do not equal the overall records reviewed (5741) because some records had missing laboratory results.

between HBV and HIV-1/2 were 0.8% among the screened blood donations.

For HCV-related coinfections, both HCV/HIV-1/2 and HCV/syphilis combinations were recorded, with seroprevalences of 1.9% and 2.0%, respectively. The proportion of coinfections among screened donations for syphilis/HIV-1/2 was 6.7%. Overall, apart from these dual infections, no triple or quadruple infections were observed.

3.5 | Temporal Trend and Seroprevalence of Transfusion-Transmissible Infection Among Blood Donors: A 5-Year Analysis

The overall seroprevalence of TTIs in the study was 18.2%. Trends in the seroprevalence of HBV, HCV, HIV-1/2, and syphilis between the years 2018 and 2024 are shown in Figure 3. Data from 2019 was not complete and was omitted in this this trend analysis. A steady decline was observed in HBV seroprevalence, which initially dropped significantly from 15.8% in 2018% to 7.1% in 2021. However, a gradual increasing trend was observed in the ensuing years from 11.0% in 2022 to 11.5% in 2023, followed by a decline in 2024 (8.3%). HCV seroprevalence was unstable, fluctuating between 5.1% and 8.1% over the period. The trend of HIV-1/2 seroprevalence across the years showed a low and decreasing seroprevalence over the period. However, a sudden increase in HIV-1/2 seroprevalence was observed in 2023 compared to the preceding years (Figure 3). The seroprevalence of syphilis was high (8.0%) in 2018. However, a steady decline was observed in the ensuing years with a temporal increase in 2022.

The trends observed for HBV, HCV, and syphilis also overlap with the recently passed COVID-19 pandemic (2020–2021) (Figure 3). A noticeable decline in seroprevalence was observed

23988835, 2026, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/hsr2.72122 by EBMG ACCESS - GHANA, Wiley Online Library on [23/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

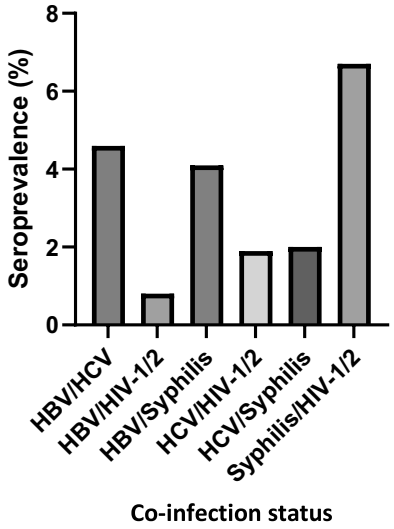


FIGURE 2 | Distribution of transfusion-transmissible infection co-infections among screened blood donations. The figure shows the percentage of blood donors who tested positive for a combination of TTIs. Percentages are calculated based on the total number of blood donations within each group. HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus.

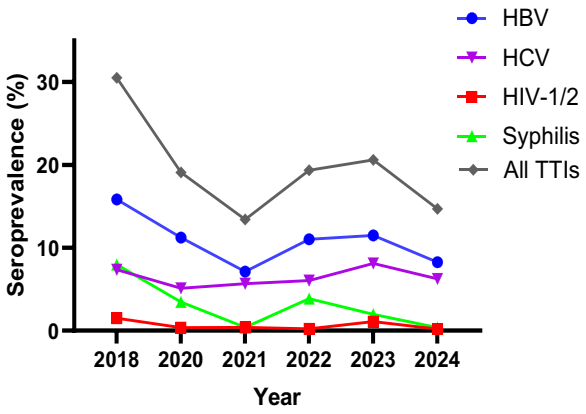


FIGURE 3 | Trend and seroprevalence of TTIs between 2018 and 2024. The figure shows the percentage (%) of blood donors who tested positive for HBV, HCV, HIV-1/2, syphilis, and overall TTIs by each year. Percentages are calculated based on the total number of blood donors within each year group. HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus, TTIs: Transfusion-transmissible infections.

to coincide with the peak years of the COVID-19 pandemic (2020–2021), when the donor recruitment numbers were also low (Table 1), followed by a gradual increase in 2022 and 2023.

4 | Discussion

The seroprevalence of TTIs among blood donors in this study was 18.2%, a finding that aligns with other local and regional reports from Ghana. For instance, a recent multicentre retrospective study across four hospitals in Ghana observed an overall TTI seroprevalence of 21.0%, with HBV (6.6%), HCV (4.9%), HIV (2.9%), and syphilis (*T. pallidum* infection) (6.8%) contributing significantly to the burden [5]. Similarly, a 5-year trend analysis at a regional hospital in Northern Ghana (2017–2022) reported a mean seroprevalence of 16.1%, with yearly fluctuations ranging from 9.5% to 31.4% [4]. These figures are consistent with our findings and emphasize the persistent public health challenge TTIs pose to blood transfusion services in resource-limited settings. At a broader scale, systematic reviews from sub-Saharan Africa have reported pooled TTI seroprevalence estimates of around 10.1%, with HBV (6.9%), HCV (2.0%), HIV (3.2%), and syphilis (2.8%) as the major contributors [35]. Compared with these regional estimates, the seroprevalence observed in our study is somewhat higher but still within the range commonly reported in Ghana. Variations in the seroprevalence across studies might be explained by the differences in the donor populations (voluntary vs. replacement donor), screening methods (rapid diagnostic test *vs.* ELISA or nucleic acid amplification tests), and geographic variations in infection burden. On the whole, the consistency of our findings with both single- and multicentre studies highlights the ongoing risk of TTIs in Ghana's blood transfusion system. It underscores the need for more sensitive screening methods, expanded donor education, and strategies to recruit low-risk voluntary donors as a means to reduce transmission risks.

The overall seroprevalence of HBV (10.1%) and HCV (6.5%) among blood donors in this study is consistent with recent Ghanaian and regional reports that classify the country as a high-endemic area for viral hepatitis. Multicentre reviews and meta-analyses from Ghana have documented HBV seroprevalence between 8% and 12% and HCV ranging from 3% to 7% among donors, reinforcing the magnitude of the transfusion risk [36]. The higher seroprevalence in male donors compared to females mirrors findings from other Ghanaian hospital-based studies, where differences have been attributed to behavioral exposures, lower vaccine uptake, and male dominance in the donor pool [37–39]. A male predominance was observed in our donor population, consistent with patterns seen in many settings, where women are underrepresented in blood donations due in part to higher deferral rates related to lower hemoglobin, physiological factors such as menstruation and pregnancy and sociocultural factors. Studies have shown that women experience significantly more deferrals for low hemoglobin compared with men and that anemia is a common reason for exclusion of female donors, particularly in sub-Saharan African context, including Ghana [40]. Age-specific patterns were also evident: HBV peaked in the younger 20–30-year-old group, while HCV seroprevalence was higher in older donors (> 50 years). These trends are well-documented in Ghana and elsewhere, with HBV reflecting early-life or young-adult transmission and HCV accumulating with age due to historical bloodborne exposures [36, 41]. These findings together highlight the persistent challenge of TTIs in resource-limited settings and the importance of strengthening donor selection, adopting more sensitive screening assays, and scaling up HBV vaccination and HCV case-finding as recommended by the World Health Organization [42]. Although ABO and Rh blood group distributions were analysed in relation to HBV and HCV seroprevalence, no statistical associations or consistent trends were observed in this study. This analysis was included as an exploratory component, given previous reports suggesting possible links between ABO blood groups and susceptibility to certain viral infections [43].

23988835, 2026, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/hsr2.72122 by EBMG ACCESS - GHANA, Wiley Online Library on [23/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

However, the current findings align with several studies indicating that ABO blood group status does not appear to be a strong or reliable predictor of TTI risk [44, 45]. Therefore, the observed patterns should be interpreted as descriptive rather than indicative of biological or clinical significance.

The present study found an overall HIV seroprevalence of 0.5% among blood donors, with a clear increase by age from 0.2% in donors aged 31–40 years to 2.8% in those above 50 years after an initial decline was observed to be 1.1% in donors below 20 years. This age-associated gradient is consistent with reports from Mali, where HIV seroprevalence among donors increased from 1.9% in the 18–30-year-old group to 3.4% in individuals over 50-year-old [46]. Similarly, a Northern Ghana donor study reported higher HIV seroprevalence in adults aged 20–49 years, further supporting the observation that older age is linked with greater risk [47]. The rise in seroprevalence with age might reflect cumulative lifetime exposure to infection risk, lower likelihood of prior screening, and possible behavioral factors. Syphilis seroprevalence in our study was 2.1%, ranging from 1.1% in younger donors (< 20 years) to 2.7% in those over 50 years. This age-dependent pattern is well-documented in sub-Saharan Africa. In Tanzania, syphilis seroprevalence among blood donors increased significantly with age [48], while studies in Ghana have shown syphilis infection to peak in middle-aged and older donors, with rates as high as 16.2% in the 40–49-year-old group [4]. Such findings underscore the persistent risk of syphilis in older donor populations, possibly linked to cumulative sexual exposure or untreated latent infections that remain detectable over time. Although the overall HIV and syphilis seroprevalence observed in this study is lower than reported in several other African donor populations, the age-related trends remain strikingly consistent. This highlights the importance of continuous age-stratified surveillance to ensure blood safety. Older donors, while often seen as more stable donors, might still represent a group requiring careful screening for TTIs. The presence of a combination of coinfections of TTIs was observed in the study. These coinfections observed in this study are consistent with previous reports from Ghana and other sub-Saharan African settings, where overlapping transmission routes contribute to multiple infections among donors [13]. Their presence underscores the need for sustained blood donation screening and integrated surveillance systems to further improve transfusion safety.

To the best of our knowledge, this study is the first long-term (5-year) trend analysis of the seroprevalence of TTIs in the Upper East Region of Ghana. The temporal fluctuations in HBV, HCV, HIV, and syphilis seroprevalence observed in this study likely reflect both underlying epidemiological patterns and operational factors. For HBV, the decline between 2018 and 2021 followed by a rebound in 2022–2023 mirrors trends reported elsewhere, where pandemic-related disruptions to blood collection changed the donor pool composition and temporarily reduced the number of high-risk donors available for screening [49, 50]. Similarly, the modest rise in HCV seroprevalence in 2023 might be attributable to recruitment of first-time or replacement donors, who are consistently shown to carry a higher TTI burden than repeated voluntary donors [51]. However, donor status (first-time vs repeat/returning) was not available in the study dataset, preventing stratified analysis by donor category. Given the known differences in TTI risk between these groups, this could be a potential source of unmeasured heterogeneity in TTI risk estimates. HIV seroprevalence remained low throughout the study period, consistent with national donor surveillance data, although small year-to-year changes likely reflect variations in donor numbers rather than true incidence shifts. For syphilis, the sharp decline after 2018 followed by small rebounds might be explained by differences in diagnostic algorithms; treponemal and nontreponemal assays have variable sensitivity and specificity, and switching between them can create apparent changes in seroprevalence unrelated to the actual infection levels [52, 53].

The observed trends in this study may, in part, reflect variations in test performance rather than true epidemiological changes. Unfortunately, detailed information on the specific RDT brands used was not available in the source records, which prevented us from reporting kit-specific performance characteristics. Additionally, 2019 represents a critical transition period immediately preceding the COVID-19 pandemic. However, some data were missing, which may affect the continuity of pre-pandemic trends and limit the ability to clearly distinguish baseline patterns from pandemic-related disruptions observed in subsequent years. While the overall long-term trends are informative, caution is required when interpreting abrupt changes around the 2018 and 2020 periods. Part of this variation may be attributable to the unavailable 2019 observations rather than a true epidemiological shift. Overall, these findings emphasize the importance of considering donor type, testing algorithms, and potential external factors such as the COVID-19 pandemic when interpreting our findings [50, 54, 55]. Understanding these multifaceted effects is essential for designing resilient blood safety strategies that can withstand future public health emergencies.

Study limitations: This study has some limitations that should be acknowledged. First, its retrospective design relied on routinely collected blood donor records, which might have been incomplete or inconsistently documented, introducing the risk of information bias. In addition, the study omitted data for 2019 due to incomplete records for that year. This might have introduced minor gaps in assessing yearly trends and could slightly affect the interpretation of temporal patterns. Second, because the data were drawn from a single primary care hospital, the findings may not be generalizable to the wider donor population in the Upper East Region or Ghana as a whole. In addition, the donor population was male-dominated, and this sex imbalance may limit the generalizability of the findings to females and affect prevalence estimates. Third, the screening of donors for TTIs was based on serological tests and rapid diagnostic test kits, which, although widely used in resource-limited settings, have lower sensitivity and specificity than ELISA or nucleic acid testing. This could have led to under- or overestimation of the true seroprevalence. In addition, donor records included only basic demographic information and did not capture detailed risk factors such as vaccination history, sexual behavior, prior transfusion, or intravenous drug use. This limited the ability to explore predictors of infection and to adjust for potential confounders such as age, year, and sex. Furthermore, while the study examines trends over 5 years, its cross-sectional nature prevents conclusions about causality or the impact of specific interventions such as vaccination campaigns or health education. Finally, the predominance of

23988835, 2026, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/hsr2.72122 by EBMG ACCESS - GHANA, Wiley Online Library on [23/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

replacement and family donors in Ghana may introduce selection bias, as this group may not reflect the general population. Despite these limitations, the study provides valuable insights into the seroprevalence and trends of HBV, HCV, HIV, and syphilis infections among blood donors in a primary health care setting in northern Ghana and highlights the ongoing need for enhanced donor screening and broader public health measures.

Conclusion: This retrospective study demonstrates that hepatitis B and C infections remain a significant concern among blood donors in the primary health care setting in the Upper East Region of Ghana. These findings supplement 5 years of trend information in a resource-constrained environment. Despite the general observation of decline in HBV and HCV over the past 5 years, the fluctuations in seroprevalence observed call for strengthening of the screening protocols and surveillance. The observation that HIV-1/2 and syphilis seroprevalence were as low is comforting, but we need to continue preventive efforts to maintain the progress. To further lower the threat of transfusion-transmissible infections and safeguard a safer blood supply in the Upper East Region of Ghana and beyond, improved criteria for donor selection, implementation of stricter blood safety protocols, and public awareness are important.

Author Contributions

Conceptualized and designed study: Sylvester Donne Dassah and Auphedeous Yinme Dang-I. Data retrieval: Courage Honorfe, Ramzy Bugri Anafo, Samuel Gana Kolog, Francis Kwame Ayimavor, and Evans Anuah. Data analysis: Francis Yennube Duut and Sylvester Donne Dassah. Writing draft manuscript: Francis Yennube Duut and Sylvester Donne Dassah. Review, editing, and approval of final version of manuscript: Francis Yennube Duut, Samuel Gana Kolog, Francis Kwame Ayimavor, Evans Anuah, Courage Honorfe, Ramzy Bugri Anafo, Godfred Agongo, James Abugri, Auphedeous Yinme Dang-I, and Sylvester Donne Dassah.

Acknowledgments

We are grateful to the management of the War Memorial Hospital for granting us access to the facility records to support the implementation of this study. We are also grateful to the staff of the hospital laboratory for their assistance during the record retrieval and data abstraction process.

Funding

The authors received no specific funding for this work.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data generated or analysed during this study are included in this published article [and its supplementary information files]. Any other information regarding the dataset used and/or analysed during this study is available from the corresponding author on reasonable request.

All authors have read and approved the final version of the manuscript. The corresponding author has full access to all of the data in this study and takes complete responsibility for the integrity of the data and the accuracy of the data analysis.

Transparency Statement

The lead author Sylvester Donne Dassah affirms that this manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

References

1. World Health Organization. Blood Safety and Availability. Geneva, Switzerland, WHO; 2025.

2. S. Puerto-Meredith, E. Singogo, M. Chagomerana, et al., "Systematic Review of Prevalence and Risk Factors of Transfusion Transmissible Infections Among Blood Donors, and Blood Safety Improvements in Southern Africa," *Transfusion Medicine* 33, no. 5 (2023): 355–371.

3. C. Mangala, D. Maulot-Bangola, A. Moutsinga, et al., "Prevalence and Factors Associated With Transfusion-Transmissible Infections (HIV, HBV, HCV and Syphilis) Among Blood Donors in Gabon: Systematic Review and Meta-Analysis," *PLoS ONE* 19, no. 8 (2024): e0307101.

4. P. Y. Hadfield, G. A. Vechey, E. Bansah, M. Nyahe, N. Khuzwayo, and E. E. Tarkang, "Transfusion-Transmissible Infections Among Blood Donors in a Regional Hospital in Ghana: A 6-Year Trend Analysis (2017–2022)," *Journal of the International Association of Providers of AIDS Care (JIAPAC)* 23 (2024), https://doi.org/10.1177/23259582241274305/ASSET/408089ED-0D16-4005-B672-4AB9977BD8B1/ASSETS/IMAGES/LARGE/.

5. W. Walana, E. K. Vicar, E. D. Kuugbee, et al., "Transfusion Transmissible Infections Among Blood Donors in Ghana: A 3-year Multicentered Health Facility-Based Retrospective Study," *Health Science Reports* 6, no. 11 (2023): e1681, https://doi.org/10.1002/HSR2.1681.

6. I. Habibu, B. M. Abubakar, I. M. Moi, and R. Abdulrazaq, "Seroprevalence of HIV, HBV, HCV and Syphilis Among Blood Donors in a Nigerian Tertiary Medical Centre," *BMC Infectious Diseases* 25, no. 1 (2025): 638.

7. D. S. Gebreyes, K. Kifetew, A. Gizaw, et al., "Prevalence and Risk Factors of Transfusion-Transmissible Infections Among Voluntary Blood Donors in North Shoa, Amhara Region, Ethiopia: A Cross-Sectional Study," *Health Science Reports* 8, no. 5 (2025): e70769.

8. World Health Organization. Screening for Transfusion-transmissible Infections. Geneva, Switzerland WHO: 2009.

9. J. P. Allain, D. Candotti, K. Soldan, et al., "The Risk of Hepatitis B Virus Infection by Transfusion in Kumasi, Ghana," *Blood* 101, no. 6 (2003): 2419–2425.

10. W. K. B. A. Owiredu, J. Osei-Yeboah, N. Amidu, and E. F. Laing, "Residual Risk of Transmission of Hepatitis B Virus Through Blood Transfusion in Ghana: Evaluation of the Performance of Rapid Immunochromatographic Assay With Enzyme Linked Immunosorbent Assay," *Journal of Medical and Biomedical Sciences* 1, no. 2 (2012): 17–28, https://www.ajol.info/index.php/jmbs/article/view/77549.

11. A. S. Alshrari, S. A. Hudu, H. Shinkafi, A. Tahir, H. Y. Raji, and A. O. Jimoh, "Prevalence and Transfusion Risks of Occult Hepatitis B Infection Among HBcAb-Positive Blood Donors in a High-Endemic Region," *Diagnostics* 15, no. 4 (2025): 486.

12. C. de Mendoza, J. M. Bautista, S. Pérez-Benavente, et al., "Screening for Retroviruses and Hepatitis Viruses Using Dried Blood Spots Reveals a High Prevalence of Occult Hepatitis B in Ghana," *Therapeutic Advances in Infectious Disease* 6 (2019): 1–8, https://doi.org/10.1177/2049936119851464/ASSET/BD41C731-DF56-434D-8BE0-866FBA37E2B1/ASSETS/IMAGES/LARGE/.

13. R. Ofori-Asenso and A. A. Agyeman, "Hepatitis B in Ghana: A Systematic Review & Meta-Analysis of Prevalence Studies (1995–2015)," *BMC Infectious Diseases* 16, no. 130 (2016), https://doi.org/10.1186/s12879-016-1467-5.

23988835, 2026, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/hsr2.72122 by EBMG ACCESS - GHANA, Wiley Online Library on [23/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

14. T. Hambridge, Y. Nartey, A. Duah, and A. Plymoth, "Hepatitis B Mother-To-Child Transmission in the Eastern Region of Ghana: A Cross-Sectional Pilot Study," *Pan African Medical Journal* 33 (2019): 218.

15. C. A. Adjei, G. T. Nachinab, F. Atibila, M. Ansah-Nyarko, J. M. Kyei, and P. K. Fosu, "Determinants and Preventive Practices of Midwives and Physicians Toward Vertical Transmission of Hepatitis B in Ghana: A Cross-Sectional Survey," *Pan African Medical Journal* 43, no. 1 (2022): 183.

16. World Health Organization. Global Health Sector Strategy on Viral Hepatitis 2016–2021. Towards Ending Viral Hepatitis. Geneva, Switzerland, WHO; 2016.

17. F. Sarkodie, M. Adarkwa, Y. Adu-Sarkodie, D. Candotti, J. W. Acheampong, and J. P. Allain, "Screening for Viral Markers in Volunteer and Replacement Blood Donors in West Africa," *Vox Sanguinis* 80, no. 3 (2001): 142–147.

18. J. T. Dongdem, et al., "Prevalence of Hepatitis B Virus Infection Among Blood Donors at the Tamale Teaching Hospital, Ghana," *BMC Research Notes* 2009, no. 115 (2012), https://doi.org/10.1186/1756-0500-5-115.

19. A. A. Adjei, H. B. Armah, F. Gbagbo, et al., "Prevalence of Human Immunodeficiency Virus, Hepatitis B Virus, Hepatitis C Virus and Syphilis Among Prison Inmates and Officers at Nsawam and Accra, Ghana," *Journal of Medical Microbiology* 55, no. 5 (2006): 593–597.

20. W. Ampofo, N. Nii-Trebi, J. Ansah, et al., "Prevalence of Blood-Borne Infectious Diseases in Blood Donors in Ghana," *Journal of Clinical Microbiology* 40, no. 9 (2002): 3523–3525.

21. R. Ephraim, I. Donko, S. Sakyi, J. Ampong, and H. Agbodjakey, "Seroprevalence and Risk Factors of Hepatitis B and Hepatitis C Infections Among Pregnant Women in the Asante Akim North Municipality of the Ashanti Region, Ghana; A Cross Sectional Study," *African Health Sciences* 15, no. 3 (2015): 709–713.

22. F. E. A. Martinson, K. A. Weigle, R. A. Royce, D. J. Weber, C. M. Suchindran, and S. M. Lemon, "Risk Factors for Horizontal Transmission of Hepatitis B Virus in a Rural District in Ghana," *American Journal of Epidemiology* 147, no. 5 (1998): 478–487, https://doi.org/10.1093/OXFORDJOURNALS.AJE.A009474.

23. N. Amidu, W. Owiredu, O. Addai-Mensah, A. Alhassan, L. Quaye, and B. Batong, "Seroprevalence and Risk Factors for Human Immunodeficiency Virus, Hepatitis B and C Vi-Ruses Infections Among Blood Donors at the Bolgatanga Regional Hospital in Bolgatanga, Ghana," *Journal of the Ghana Science Association* 12, no. 1 (2010): 1–12, https://doi.org/10.4314/jgsa.v12i1.56794.

24. A. E. Quintas, N. Cuboia, L. Cordeiro, A. Sarmento, and L. Azevedo, "Seroprevalence of Human Immunodeficiency Virus in African Blood Donors: A Systematic Review and Meta-Analysis," *EBioMedicine* 105 (2024): 105210.

25. C. Nkansah, D. Serwaa, F. Osei-Boakye, and R. Owusu-Ampomah, "Magnitude and Trend of HIV and Treponema Pallidum Infections Among Blood Donors in Offinso-North District, Ghana: A Nine-Year Retrospective, Cross-Sectional Study," *African Health Sciences* 22, no. 1 (2022): 465–474.

26. F. Sarkodie, O. Hassall, E. Owusu-Dabo, et al., "Syphilis Screening Practices in Blood Transfusion Facilities in Ghana," *International Journal of Infectious Diseases* 43 (2016): 90–94, https://doi.org/10.1016/j.ijid.2015.12.020.

27. M. Delaney, S. Telke, S. Zou, et al., "The BLOODSAFE Program: Building the Future of Access to Safe Blood in Sub-Saharan Africa," *Transfusion* 62, no. 11 (2022): 2282–2290.

28. National Blood Service Ghana. National Strategy for Blood Screening. National Blood Service Ghana. 2022.

29. F. Baffour-Ata, P. Antwi-Agyei, G. O. Apawu, et al., "Using Traditional Agroecological Knowledge to Adapt to Climate Change and Variability in the Upper East Region of Ghana," *Environmental Challenges* 4 (2021): 100205, https://doi.org/10.1016/J.ENVC.2021.100205.

30. Ghana Statistical Service. Ghana 2021 Population and Housing Census: Population of Regions and Districts. Accra, GSS; 2021.

31. A. R. Oduro, G. Wak, D. Azongo, et al., "Profile of the Navrongo Health and Demographic Surveillance System," *International Journal of Epidemiology* 41, no. 4 (2012): 968–976.

32. E. A. Nonterah, E. Adomolga, A. Yidana, et al., "Descriptive Epidemiology of Anaemia Among Pregnant Women Initiating Antenatal Care in Rural Northern Ghana," *African Journal of Primary Health Care & Family Medicine* 11, no. 1 (2019): 1.

33. A. Amini, O. Varsaneux, H. Kelly, et al., "Diagnostic Accuracy of Tests to Detect Hepatitis B Surface Antigen: A Systematic Review of the Literature and Meta-Analysis," *BMC Infectious Diseases* 17, no. 1 (2017): 698, https://doi.org/10.1186/S12879-017-2772-3.

34. G. Jargalsaikhan, M. Eichner, D. Boldbaatar, et al., "Sensitivity and Specificity of Commercially Available Rapid Diagnostic Tests for Viral Hepatitis B and C Screening in Serum Samples," *PLoS ONE* 15, no. 7 (2020): e0235036, https://doi.org/10.1371/JOURNAL.PONE.0235036.

35. Y. A. Karikari, "Transfusion-Transmissible Infections Among Blood Donors in Sub-Saharan Africa: Systematic Review and Meta-Analysis," *University of Ghana* (2021), http://ugspace.ug.edu.gh:8080/handle/123456789/41684.

36. J. Abesig, Y. Chen, H. Wang, F. M. Sompo, and I. X. Y. Wu, "Prevalence of Viral Hepatitis B in Ghana Between 2015 and 2019: A Systematic Review and Meta-Analysis," *PLoS ONE* 15, no. 6 (2020): e0234348, https://doi.org/10.1371/JOURNAL.PONE.0234348.

37. D. Kubicka-Russel, A. Kopacz, E. Sulkowska, M. Łętowska, and P. Grabarczyk, "Demographic Changes in the Polish Blood Donors Eligible for Blood Donation and Screened for Transfusion-Transmitted Infections (2005–2018)," *Journal of Transfusion Medicine* 16, no. 2 (2023): 39–64.

38. C. Eren, "An Analysis on HBsAg, Anti-HCV, Anti-HIV½ and VDRL Test Results in Blood Donors According to Gender, Age Range and Years," *PLoS ONE* 14, no. 9 (2019): e0219709.

39. A. K. Mohamud, P. Inchon, S. Suwannaporn, K. Prasert, and N. I. Dirie, "Assessment of Prevalence and Risk Factors Associated With Hepatitis B Virus Infection Among Blood Donors in Mogadishu Somalia," *BMC Public Health* 24, no. 1 (2024): 690.

40. A. E. Mast, K. S. Schlumpf, D. J. Wright, et al., "Demographic Correlates of Low Hemoglobin Deferral Among Prospective Whole Blood Donors," *TRANSFUSION* 50, no. 8 (2010): 1794–1802, https://doi.org/10.1111/J.1537-2995.2010.02649.X.

41. A. M. Nlankpe, P. K. Kwadzokpui, D. Adedia, J. N. Nignan, and P. K. Owiafe, "Seroepidemiology of Hepatitis B and C Virus Infections: A Five-Year Retrospective Study Among Blood Donors in Saboba District in the Northern Region of Ghana," *BioMed Research International* 2021 (2021): 5599705.

42. World Health Organization. Hepatitis B. Geneva, Switzerland, WHO; 2025.

43. W. Jing, S. Zhao, J. Liu, and M. Liu, "ABO Blood Groups and Hepatitis B Virus Infection: A Systematic Review and Meta-Analysis," *BMJ Open* 10, no. 1 (2020): e034114, https://doi.org/10.1136/BMJOPEN-2019-034114.

44. B. Legese, M. Shiferaw, W. Tamir, et al., "Association of ABO and Rhesus Blood Types With Transfusion-Transmitted Infections (TTIs) Among Apparently Healthy Blood Donors at Bahir Dar Blood Bank, Bahir Dar, North West, Ethiopia: A Retrospective Cross-Sectional

23988835, 2026, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/hsr2.72122 by EBMG ACCESS - GHANA, Wiley Online Library on [23/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Study," *Journal of Blood Medicine* 13 (2022): 581–587, https://doi.org/10.2147/JBM.S374851.

45. R. Behal, R. Jain, K. K. Behal, and T. N. Dhole, "Variation in the Host ABO Blood Group May Be Associated With Susceptibility to Hepatitis C Virus Infection," *Epidemiology and Infection* 138, no. 8 (2010): 1096–1099, https://doi.org/10.1017/S0950268809991117.

46. A. Jary, S. Dienta, V. Leducq, et al., "Seroprevalence and Risk Factors for HIV, HCV, HBV and Syphilis Among Blood Donors in Mali," *BMC Infectious Diseases* 19, no. 1 (2019): 1064.

47. A. Tikumah Muktar, A. Yidana, M. Wombeogo, and E. Weyori Weikam, "Prevalence and Trends of HIV and Syphilis Among Blood Donors in the Northern Region of Ghana; A Five-Year Retrospective Study," *World Journal of Public Health* 4, no. 3 (2019): 65.

48. M. I. Matee, P. M. Magesa, and E. F. Lyamuya, "Seroprevalence of Human Immunodeficiency Virus, Hepatitis B and C Viruses and Syphilis Infections Among Blood Donors at the Muhimbili National Hospital in Dar Es Salaam, Tanzania," *BMC Public Health* 6, no. 1 (2006): 21, https://doi.org/10.1186/1471-2458-6-21/TABLES/2.

49. K. B. David, K. Simfukwe, M. B. Musa, S. Munharo, and D. E. Lucero-Prisno, "Impact of COVID-19 on Blood Donation and Supply in Africa," *African Journal of Laboratory Medicine* 10, no. 1 (2021): 1408.

50. N. Kukar, A. Handa, N. Syal, P. Garg, N. S. Gopalpuri, and H. Kaur, "Trends of Coinfections Among Healthy Blood Donors: COVID-19 Pandemic Repercussion," *Journal of Family Medicine and Primary Care* 13, no. 10 (2024): 4394–4398, https://doi.org/10.4103/JFMPC.JFMPC_120_24.

51. S. Puerto-Meredith, E. Singogo, M. Chagomerana, et al., "Systematic Review of Prevalence and Risk Factors of Transfusion Transmissible Infections Among Blood Donors, and Blood Safety Improvements in Southern Africa," *Transfusion Medicine* 33, no. 5 (2023): 355–371.

52. J. R. Papp, I. U. Park, Y. Fakile, L. Pereira, A. Pillay, and G. A. Bolan, "CDC Laboratory Recommendations for Syphilis Testing, United States, 2024," *MMWR: Recommendations and Reports* 73, no. 1 (2024): 1–32.

53. A. M. Al-Matary and F. A. S. A. Gashaa, "Comparison of Different Rapid Screening Tests and ELISA for HBV, HCV, and HIV Among Healthy Blood Donors and Recipients at Jibla University Hospital Yemen," *Journal of Medicine and Life* 15, no. 11 (2022): 1403–1408.

54. Y. A. Nartey, R. Okine, A. Seake-Kwawu, et al., "A Nationwide Cross-Sectional Review of In-Hospital Hepatitis B Virus Testing and Disease Burden Estimation in Ghana, 2016−2021," *BMC Public Health* 22, no. 1 (2022): 2149, https://doi.org/10.1186/S12889-022-14618-3.

55. Y. A. Nartey, R. Okine, A. Seake-Kwawu, et al., "Hepatitis C Virus Seroprevalence, Testing, and Treatment Capacity in Public Health Facilities in Ghana, 2016–2021; A Multi-Centre Cross-Sectional Study," *PLoS ONE* 18, no. 6 (2023): e0287580, https://doi.org/10.1371/JOURNAL.PONE.0287580.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.
Supplementary_Figure_1.

23988835, 2026, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/hsr2.72122 by EBMG ACCESS - GHANA, Wiley Online Library on [23/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License